

Citation Verification Protocol

(CVP)

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Abstract

This document establishes the Citation Verification Protocol (CVP), a domain-general method for verifying citations in any document whose claims rest on cited sources. CVP ensures that every citation is (1) real and correctly identified and (2) genuinely supports the specific claim it is used to justify. The second requirement is what distinguishes CVP from ordinary citation checking. The verifier must retrieve the source, read the portion of it that bears on the claim, and record the evidence found there, rather than confirming only that the work exists. CVP applies wherever a fabricated or misattributed citation carries a cost, including academic manuscripts, legal filings, clinical and regulatory documents, analyst and consulting reports, and internal business research. The protocol is designed to be executed using AI systems with live browsing, while producing an auditable verification log that can be independently reviewed.

1. Purpose and Scope

This protocol provides a systematic method for verifying citations used in any document that relies on them. It was developed for Synthience Institute manuscripts and research artifacts and is published for general use. Nothing in the three-part verification method of Section 5 is specific to academic publishing: the operation of checking whether a source substantiates the use it is being put to is the same whether the source is a journal article, a court opinion, a clinical guideline, a technical standard, or an internal report.

CVP addresses two verification requirements:

- **Existence verification:** Confirming that cited works exist and match the cited bibliographic identity.
- **Support verification:** Confirming that cited works actually support the specific claims they are cited to justify.

Critical requirement: This protocol is only effective when the verifier retrieves the actual source text at verification time, whether by live browsing, licensed database access, or a lawfully held copy.

Prior-based assessments (model memory, training priors, general plausibility judgments) are explicitly insufficient, and no support rating may be assigned from an abstract, an index record, a snippet, or a summary. Access may be restricted. Reading may not be skipped. This requirement is absolute and does not vary by domain or by profile.

1.1 Public Verifiability Constraint (PVC): Access Gate and Profiles

The Public Verifiability Constraint governs what counts as adequate access to a source. It operates through one of two profiles, selected before verification begins and recorded in the log. The profile determines whether restricted access disqualifies a citation. It never relaxes the requirement that the verifier read the portion of the source that bears on the claim.

Profile 1, Open-Publication. The default for Synthience Institute publications, and for any document whose readers must be able to check its sources without payment. Sources must be publicly accessible for free at the time of verification. This is a hard exclusion rule: if a source is paywalled, requires authentication, is only partially viewable, or cannot be retrieved in full, it must not appear in the references.

Profile 2, Licensed-Access. For legal, clinical, regulatory, commercial, and internal work, where the authoritative sources are licensed, purchased, or unpublished. Access Class is recorded descriptively rather than as an exclusion gate. A citation passes if the verifier held lawful access to the complete text at verification time, the log records the basis of that access, and the log states plainly whether an independent reader can reach the source. Restricted access is disclosed, not disqualifying.

What does not change between profiles. The verifier must reach and read the portion of the source that bears on the claim. A citation whose relevant text could not be read, under either profile, cannot receive a FULL SUPPORT rating. Profile 2 widens what may be cited. It does not widen what may be asserted about a source that was never opened.

Access Class (required for every citation)

PASS

- **FREE-FULLTEXT:** Complete text accessible without payment or authentication at the time of verification. Passes under both profiles.

FAIL

- **PAYWALLED:** Payment required and not held. Fails under both profiles. A paywalled source the verifier has lawfully purchased or licensed is classed LICENSED-FULLTEXT; this class covers only sources the verifier could not reach.
- **AUTH-REQUIRED:** Institutional login or account required and not held. Fails under both profiles.
- **PARTIAL-ONLY:** Abstract only, preview only, limited pages, "snippet view," or any form of incomplete access. Fails under both profiles, without exception. This is the class that most often produces a false pass, because a plausible abstract invites a support rating the full text would not sustain.
- **UNAVAILABLE:** Link rot, removed content, dead URL, or otherwise not retrievable.
- **LICENSED-FULLTEXT (Profile 2 only):** Complete text accessible to the verifier under a subscription, institutional licence, or purchase, including a copy lawfully exported for verification. Record the platform and the basis of access.
- **INTERNAL (Profile 2 only):** Complete text held lawfully by the verifying organisation and not publicly published. Record the document identifier, version, and custodian.

Clarification: What counts as FREE-FULLTEXT

FREE-FULLTEXT includes any format that allows verification of the full work, including:

- Publisher-hosted HTML full text (even if PDF is blocked), provided the full text is complete.
- Preprint repositories or archives (arXiv, institutional repositories) with full text.
- Author-accepted manuscripts (AAM) or equivalent full manuscripts, provided they are complete and untruncated.

If the verifier cannot access the complete text needed to check claim support, the citation fails PVC.

1.1.1 Foundational-Text Provision (Established Works Not Retrieved)

A narrow provision covers long-established works, typically books, that are cited for the proposition they are widely known for, and whose full text the verifier did not obtain. It exists because such works were in practice being passed on convergent citation rather than on retrieval, and an unwritten allowance is more dangerous than a stated one.

What it permits. Part A identity verification may be satisfied without the full text where the work is confirmable through authoritative catalogue records (publisher, ISBN, edition, library holdings) and is cited by many independent sources for the same proposition. Identity is genuinely established this way; a widely held book is not a fabrication risk.

What it does not permit. Part C claim support may not be satisfied this way. Convergent citation is evidence about what the field reports the work to say. It is not evidence from the work. A citation passed under this provision receives the support rating ATTRIBUTED, not FULL SUPPORT, and the log must record that the rating rests on convergent independent attribution rather than on the source text.

Hard limit. A citation rated ATTRIBUTED may not carry a load-bearing claim. If the document's argument depends on what the work actually says, obtain the work. The provision covers background, orientation, and canonical attribution, and nothing else. Any verifier applying it must state in the log why the claim is not load-bearing.

1.2 Persistence and Archiving Requirement (PAR)

PVC ensures free public access at verification time. Sources can later disappear or change. Therefore, CVP requires a persistence record for each verified citation.

For every citation that passes PVC, the verifier must record at least one of the following in the verification log:

- A stable canonical identifier (DOI, arXiv ID, ACL Anthology ID, OpenReview ID, PubMed ID, etc.) plus the canonical landing page URL, and
- An archival reference (public web archive snapshot URL) when available, or
- A local snapshot record (internal copy) with a content hash, stored for audit continuity.

Local snapshots do not replace PVC. They exist to preserve auditability if the public source later changes or disappears.

2. The Problem This Protocol Addresses

AI systems routinely fail citation verification in predictable ways. CVP is designed to prevent these failures.

2.1 Tool Invocation Failures

- Simulated browsing: the system claims it "checked" a source but did not open it.
- URL-free verification: the system provides conclusions without evidence links.

2.2 Completeness Failures

- Partial verification: verifying only a subset of citations while implying the entire reference list was checked.
- Denominator loss: failure to report how many citations exist versus how many were verified.

2.3 Support Verification Failures

- Existence-only validation: confirming the paper exists but not checking whether it supports the claim.
- Abstract-only validation: using abstracts as if they prove support.
- Prior-based rejection: dismissing valid citations as "fake" based on model priors rather than evidence.

3. Platform Compatibility

CVP may be executed on any AI system that can:

- Open live web pages, or access licensed databases and lawfully held local copies,
- Retrieve full-text sources,
- Report the locators actually accessed,
- Produce a structured verification log.

If the verifier cannot reach source text by any route, CVP cannot be performed. Live web browsing is the usual route and the only one available under Profile 1; under Profile 2 a licensed database session or a lawfully exported copy supplied to the verifier serves equally.

4. Pre-Verification Requirements

4.1 Confirm Verifier Has Source Access

Before verifying any document, the verifier must demonstrate working source access by opening at least one known source and recording:

- The locator opened (URL, database record, or file identifier)
- The access date and time
- A short content confirmation (one sentence) that could only be produced by actually opening the source

4.2 Create a Complete Citation Inventory

The verifier must produce an inventory of all citations used in the manuscript, including:

- Total number of citations (denominator)
- Unique identifier per citation (C1, C2, ..., CN)

- Full reference text as written
- The manuscript claim(s) each citation supports (Claim IDs)

CVP is incomplete until all citations in the inventory are processed.

4.3 Select Profile and Enforce PVC Before Verification

Select and record the PVC profile before verification begins. For each citation, determine Access Class immediately. Any citation that fails the selected profile must be removed or replaced before the document is eligible for release. Under Profile 2, a citation classed LICENSED-FULLTEXT or INTERNAL passes, and the log must state that an independent reader without the same access cannot reach it.

4.4 Claim Map Requirement (No "floating citations")

Every citation must be tied to a specific claim (or set of claims) in the manuscript. Citations not mapped to claims are invalid and must be removed or reattached with explicit claim linkage.

4.5 Citation Brief Method

By default the verifier receives the full document and works from it. For large documents or multi-document runs, the verification input may instead be a citation brief: an extract containing only what verification requires. The verifier never ingests the full document.

Why it exists. Full-document ingestion consumes verifier context, slows the run, and encourages skimming in place of verification. In a seven-document Institute run, a brief of roughly five kilobytes replaced five hundred kilobytes of ingestion, the verifier proceeded directly to opening sources, and no context pressure was observed.

Brief construction. One entry per citation, containing: the cited passage verbatim as it appears in the document; the claim the citation must support, stated as a claim; the full reference as written; and the primary URL or locator.

When to use it. Recommended for multi-document runs and for any document exceeding roughly fifty kilobytes. The full document remains acceptable, and is preferable, for single short documents where context is not under pressure.

Constraint. The brief must be built from the current version of the document, after all content revisions have been applied. A brief built from a superseded draft verifies a document that no longer exists. Sources passed under the foundational-text provision need no brief entry, since no source retrieval is performed.

4.6 Multi-Pass Escalation

A single verification pass does not always find evidence that exists. Evidence may sit in an appendix, a supplementary file, or a companion document on a different domain from the source itself. In one Institute run, a statistic required four passes before it was located in a standalone appendix hosted separately from the main paper.

Where a claim's supporting evidence cannot be located in the primary source, escalate in this order before recording a failure:

- **Step 1.** Retrieve the primary URL or locator.

- **Step 2.** Try alternative access paths for the same work: repository copies, publisher HTML where the PDF is blocked, author-hosted manuscripts, web archive snapshots.
- **Step 3.** Check for supplementary materials, appendices, datasets, or companion documents linked from the primary source. This is the step most often skipped, and the one that most often holds the missing evidence.
- **Step 4.** If steps 1 to 3 fail, run a wider search using a different platform or a deep research tool, on the reasoning that a single retriever's coverage is not the field's coverage.

Recording the result. Where evidence is found in a supplementary document rather than the main source, the reference must be updated to include the supplementary locator, so a later reader reaches the evidence rather than repeating the search. A claim is not failed merely because it does not appear in the main document: supplementary materials linked from the canonical source are legitimate evidence anchors. A claim is failed when steps 1 to 4 are exhausted and the evidence is still not found, and the log must record which steps were run.

5. Three-Part Verification Method

Each citation must pass all three parts to be retained.

Part A: Identity and Existence Verification

Confirm the cited work's identity matches the reference:

- Title, authors, year, venue
- Canonical identifier (preferred), by source type. Scholarly: DOI, arXiv, PubMed, ACL Anthology, OpenReview, ISBN. Legal: reporter citation, docket number, statute or regulation number, ECLI. Clinical and regulatory: trial registration number, guideline identifier, marketing authorisation or approval number. Standards and commercial: ISO, IEC or IEEE number; report identifier plus publisher. Internal: document identifier, version, and custodian.
- Canonical landing page URL, or the database record locator where no public page exists

Record any mismatches. If the cited identity cannot be confirmed, the citation fails.

Part B: Access Verification and Canonical Locator Priority

Determine Access Class under the PVC profile selected in Section 4.3.

Prefer canonical locators in this order (when available):

1. DOI landing page that provides FREE-FULLTEXT, or links to a FREE-FULLTEXT version
2. First-party repository or official publisher of record (arXiv, ACL Anthology, PubMed Central, institutional repository; for non-scholarly sources, the issuing court, agency, regulator, standards body, or originating organisation)
3. Author-hosted full manuscript at a stable URL, or, under Profile 2 only, a licensed database copy or lawfully held internal copy

If only secondary mirrors exist, the verifier must justify their use in the log and must still reach complete text. Where escalation beyond the primary locator was required, record the steps run per Section 4.6.

Part C: Claim Support Verification (Evidence Standard)

For each Claim ID linked to the citation:

- Extract the claim text from the manuscript (verbatim).

- Open the full text of the source (not abstract-only).
- Locate evidence anchors in the source that support the claim.

Evidence anchors must include:

- A pointer: page number(s) if PDF, or section heading(s) if HTML
- A short excerpt (maximum 25 words) or a precise paraphrase if quoting is not feasible
- A one-sentence explanation of why this evidence supports the claim

Support Rating (required):

- **FULL SUPPORT:** Evidence directly supports the claim as written.
- **PARTIAL SUPPORT:** Evidence supports only a weaker or narrower version of the claim.
- **NO SUPPORT:** Evidence does not support the claim.
- **AMBIGUOUS:** Cannot confidently determine support from the source text (treated as failure).
- **ATTRIBUTED:** The source was not read, and the rating rests on convergent independent attribution of the same proposition to the same work. Available only under the foundational-text provision of Section 1.1.1, and only for claims that are not load-bearing.

Disposition Rules:

- Only FULL SUPPORT may pass, except where ATTRIBUTED is expressly permitted under Section 1.1.1.
- PARTIAL SUPPORT requires manuscript revision (narrow the claim) or source replacement.
- NO SUPPORT requires source removal or claim removal.
- AMBIGUOUS requires source replacement or claim rework until verifiable.
- **ATTRIBUTED** passes only if the claim is not load-bearing and the log records why. If the argument depends on what the source says, obtain the source and re-rate.

6. Implementation Procedure

1. Confirm live browsing access (Section 4.1).
2. Generate complete citation inventory with denominator (Section 4.2).
3. For each citation C1...CN:
 - Part A: verify identity and capture canonical ID + URL
 - Part B: assign Access Class; enforce PVC; apply canonical locator priority
 - Part C: verify claim support with evidence anchors and rating
4. Produce a Verification Log (Section 9).
5. Produce a Certification Statement indicating:
 - Total citations: N
 - Verified citations: N (must match)
 - Passed citations: P
 - Removed/replaced citations: R
 - Date/time of verification run

7. Verification Tiers

Tiering determines how defensible the verification is under scrutiny.

- **Tier 0 (Not Verified):** No CVP log. Not publishable under Synthience Institute standards.
- **Tier 1 (Single-Platform Verified):** Each citation verified on one platform with evidence anchors.

- **Tier 2 (Dual-Platform Verified):** Each citation verified on two independent platforms or repositories when feasible.
- **Tier 3 (Audit-Ready):** Tier 2 plus persistence record (PAR) and a complete, reviewer-readable log.

Default for public release: Tier 2 minimum. Tier 3 recommended for flagship documents.

8. Failure Mode Recognition Guide

If any of the following occur, verification must be treated as failed and rerun:

- The verifier provides conclusions without URLs opened.
- The verifier cites abstracts as if they prove claim support.
- The verifier dismisses a source as "fake" without browsing evidence.
- The verifier verifies fewer than N citations but reports completion.
- The verifier cannot provide evidence anchors (page/section pointers) for support claims.
- The verifier cannot classify Access Class for a citation.

9. Documentation Requirements (Verification Log)

A CVP Verification Log must be produced for every manuscript and must include:

Manuscript metadata

- Manuscript title, version, date
- Verifier identity (human or AI platform name); verification date/time

For each citation (C#)

- Full reference text
- Canonical identifier(s) and canonical URL
- PVC profile applied, and Access Class under it
- Retrieval locator(s) actually opened, and the basis of access where not public
- Claim IDs supported

For each claim

Claim text (verbatim)
Evidence anchor pointer (page/section)
Short excerpt (≤ 25 words) or precise paraphrase
Support Rating (FULL / PARTIAL / NONE / AMBIGUOUS / ATTRIBUTED)
Disposition (KEEP / REVISE CLAIM / REPLACE SOURCE / REMOVE)

Persistence and Archiving Requirement (PAR) fields

- Access date and time for the full text, and the escalation steps run if any
- Archival URL if created or found
- Local snapshot hash (if stored internally), plus storage location reference

Certification Statement Template

I certify that I executed SF0037 CVP v1.5, retrieving and reading the portion of each cited source that bears on the claim it supports. PVC profile applied: _____ (Open-Publication / Licensed-Access). Verification input: _____ (full document / citation brief). Total citations inventoried: N = ____

Citations fully processed: ____ / ____
Citations retained after PVC and support verification: ____
Citations removed or replaced: ____
Verification tier achieved: Tier ____
Verification completed on: _____

Appendix A: Quick Reference Checklist

- Source access confirmed (locator + timestamp recorded)
- Citation inventory complete (denominator N present)
- PVC profile selected and recorded; Access Class recorded for every citation
- Canonical identifier recorded, appropriate to source type
- Full text opened for every citation
- Each citation mapped to Claim ID(s)
- Evidence anchors recorded for every claim (page/section + excerpt/paraphrase)
- Support rating applied (FULL required to pass, except ATTRIBUTED under Section 1.1.1)
- Persistence record captured (archival link and/or local hash)
- Certification statement completed
- Escalation steps recorded where the primary locator did not yield the evidence
- Citation brief, if used, built from the current version of the document

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